

EFMW-EXP-001-v1.0

Recursive Coherence Early-Warning Benchmark

Status: FROZEN BEFORE CONFIRMATORY EXECUTION

Primary Claim

On the frozen synthetic partially observed control benchmark, the EFMW-style recursive coherence monitor must achieve a median paired warning-lead gain of at least 10 timesteps over the conventional residual EWMA monitor.

Scope Limitation

This experiment tests a bounded recursive monitoring mechanism on a synthetic benchmark. It does not validate EFMW physics generally or establish the canonical EFMW field equation.

Frozen Monitor Definitions

Baseline: $B_t = 0.90 B_{(t-1)} + 0.10 |r_t|$; threshold = 0.0455502027203.

EFMW-style: $M_t = 0.97 M_{(t-1)} + 0.03 r_t$; $C_t = |M_t|$; threshold = 0.00834791535744.

Alert: first threshold exceedance sustained for three consecutive samples.

Frozen Confirmatory Predictions

ID	Endpoint	Frozen success rule
P001	paired_lead_gain	median(EFMW lead - baseline lead) $\geq$ 10 timesteps
P002	efmw_median_lead_time	median EFMW warning lead $\geq$ 55 timesteps
P003	efmw_raw_false_positive_rate	$0.03 \leq$ EFMW raw pre-onset FPR $\leq$ 0.07
P004	baseline_raw_false_positive_rate	$0.03 \leq$ baseline raw pre-onset FPR $\leq$ 0.07
P005	early_warning_win_rate	fraction of seeds with EFMW lead $>$ baseline lead $\geq$ 0.60
P006	compute_ratio	median EFMW monitor runtime / baseline monitor runtime $\leq$ 1.50

Freeze Rules

Theory, thresholds, seeds, endpoints, scoring rules, exclusions, and ablations may not be changed after freeze. Any correction requires a new packet version. Confirmatory failure remains a failure of this frozen formulation.

Governing Principle

Prediction before observation. Ablation before interpretation. Replication before generalization.